

Data

Table of contents

Our Dataset	1
Why did we choose this dataset?	1
Source	1
Data Preparation	1
Download the cleaned data	2
Schema at a glance	2
A peek at the data	2

Our Dataset

The dataset captures **hazardous cargo (HNS, Hazardous and Noxious Substances) handled at Turkish ports during 2022**. It was published by the **Directorate General for Maritime Affairs** (T.C. Ulaştırma ve Altyapı Bakanlığı – Denizcilik Genel Müdürlüğü).

Each leaf record describes a single cargo movement at the resolution of:

- the **regional port authority** (e.g., *Aliağa Bölge Liman Başkanlığı*),
- the **facility** that handled the cargo (e.g., *Egegaz Aliağa LNG Terminal*),
- the **receiver / final consignee company**,
- the **cargo amount in metric tons**.

Why did we choose this dataset?

We chose it for three reasons:

1. **It's real, operational, and Turkish.** The patterns we surface are about the actual industrial geography of Türkiye's coastlines.
2. **It supports policy questions.** The Maritime Affairs office has flagged firms handling more than 20,000 tons of hazardous cargo as candidates for upcoming tax/regulatory measures. The dataset lets us put numbers on that conversation.
3. **It's multi-dimensional.** Company $\times$ facility $\times$ port authority gives us room for descriptive, geographic, and threshold-based analyses without leaving a single source.

Source

Directorate General for Maritime Affairs. *2022 Hazardous Cargo Handling Records*. Internal compilation, accessed via project supervisor.

The original Excel workbook is bundled with this site as `data/HNS_2022.xlsx`, alongside the cleaned versions described below.

Data Preparation

The raw workbook is a pivot-table export with three nested categorical levels (port authority $\rightarrow$ facility $\rightarrow$ company) and built-in subtotal rows ("Toplam ..."). To analyse it we need a flat, tidy table.

The initial set of data collected from Directorate General for Maritime Affairs was not in an appropriate condition for analysis. The initial data had nested categories, subtotals, inconsistent names and missing values which should be considered while processing the data.

In order to make the data analysis process more appropriate in R, the dataset underwent a pre-processing process. First, unneeded subtotals were omitted from the dataset to ensure no double calculations. In addition, rows without the name of the companies and cargos were excluded from further analysis.

An extremely critical step in the pre-processing phase was cleaning company names. Different types of the same name like STAR RAFİNERİ AŞ, STAR RAFİNERİ A.Ş., and Star Rafineri Aş were used in the raw dataset. If these company names were not processed, each of them would have been considered a different company in the groupings and visualization phases. This way, all of the company names were standardized to BOTAŞ.

Moreover, other text-based variables like Port Authority Name and Facility Name columns were also standardized in terms of text to simplify further processes.

Re-run the script any time the source workbook changes:

```
source("R/prepare_data.R")
```

It writes:

- `data/HNS_2022_clean.RData`, the canonical R object.
- `data/HNS_2022_clean.csv`, the same table in CSV (UTF-8) for inspection / sharing.

Download the cleaned data

Tip

The cleaned R dataset is available here: [HNS_2022_clean.RData](#) (also available as [HNS_2022_clean.csv](#)).

Loading it back is a one-liner:

```
load("data/HNS_2022_clean.RData") # creates a data frame called `hns2022`
```

Schema at a glance

Column	Type	Description
<code>port_authority</code>	character	Full port-authority name (e.g., <i>Aliğa Bölge Liman Başkanlığı</i>)
<code>port_authority_short</code>	character	Short form (e.g., <i>Aliğa</i>), used for joins
<code>facility</code>	character	Facility / terminal that handled the cargo
<code>company</code>	character	Receiver / consignee, as written in the source
<code>company_clean</code>	character	Normalised company name for grouping near-duplicates
<code>cargo_mt</code>	numeric	Cargo amount, in metric tons
<code>province</code>	character	Host province of the port authority
<code>iso2</code>	character	ISO 3166-2 code for the province (e.g., TR-35)

A peek at the data

```
hns2022 |>
  dplyr::select(port_authority_short, facility, company, cargo_mt, province) |>
```

```

head(10) |>
kable(
  col.names = c("Port authority", "Facility", "Company", "Cargo (m. ton)", "Province"),
  format.args = list(big.mark = ",")
) |>
kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)

```

Table 2: First 10 rows of the cleaned dataset.

Port authority	Facility	Company	Cargo (m. ton)	Province
Aliğa	TÜPRAŞ	İzmir Aliğa Rafinerisi	15,088,213.00	İzmir
Aliğa	STAR ALIAGA	STAR RAFİNERİ A.Ş	12,114,504.29	İzmir
Aliğa	STAR ALIAGA	SOCAR	1,173,424.00	İzmir
Aliğa	STAR ALIAGA	PETKİM	222,422.88	İzmir
Aliğa	STAR ALIAGA	POAŞ	93,866.95	İzmir
Aliğa	STAR ALIAGA	SHELL/ISTANBUL JET/SOCAR	67,807.00	İzmir
Aliğa	STAR ALIAGA	STAR	65,734.57	İzmir
Aliğa	STAR ALIAGA	OMV	49,875.76	İzmir
Aliğa	STAR ALIAGA	SOCAR,SHELL,THY OPET	49,211.41	İzmir
Aliğa	STAR ALIAGA	THY OPET	32,855.83	İzmir